

KARAKALA BHANU PRAKASH REDDY

Software Development Engineer-II | Bangalore, India

karakalabhanuprakashreddy@gmail.com | +91-7032718222 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Results-driven SDE with 3+ years building event-driven microservices at Oracle and production-grade mobility platforms at Reflex Mobility using Java, Spring Boot, Kafka, and React. Reduced latency by 550ms, cut CPU by 40%, processed 5TB/day of JSON workloads, and owned 90% of 5 production services powering corporate transportation for top enterprises across India.

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, JavaScript, C++, Shell Scripting

Frameworks: Spring Boot, Spring Security, GraphQL, JPA, Hibernate, JBPM, Angular, React, TypeScript, WebClient, Maven, Gradle

Tools: Kafka, Docker, JUnit, Karate, Cucumber, Git, CI/CD, AWS, Azure, Linux, Jira, Jenkins, WebSocket, Google Maps API

Databases: MySQL, SQL Server, Redis, Spring Cache

Integrations: Claude AI, Exotel, MoveInSync, GstZen, Rapido, Msg91, Zoho Mail, Slack API

Concepts: Microservices, REST APIs, Event-Driven Architecture, OOP, Design Patterns, Agile, Asynchronous Programming

PROFESSIONAL EXPERIENCE

Software Development Engineer-II

May 2026 – Sep 2026

Reflex Green Mobility Ltd. (RGML), Bangalore, India

- Owned 90% of 5 production services end-to-end: ride dispatch backend, enterprise ride management, enterprise dashboard (React/TypeScript), ops dashboard, and intelligent driver allocation engine (Spring Boot)
- Designed 4-stage driver allocation pipeline (Eligibility → Pool → Match → Score) with point-in-polygon geo-boundary checks and geo-distance weighted scoring, powering real-time ride assignment at scale
- Integrated Claude AI (Anthropic) with Zoho Mail for automated ride audit — parsed inbound trip emails, extracted structured data, and reconciled against booked rides, reducing manual audit effort significantly
- Built annotation-driven idempotency framework with SQL-backed deduplication preventing duplicate ride-action processing across distributed and retried requests
- Shipped GST e-invoicing with GstZen OAuth 2.0 and MoveInSync vendor trip data integration with H3 geo-cell tracking
- Integrated Exotel Exophone for virtual phone number masking per ride, protecting driver and passenger privacy
- Implemented vehicle downgrade charge waiver logic across cancellation and No Show flows in both consumer and enterprise services
- Built WebSocket-driven real-time ride monitoring with live driver location streaming pushed to all connected enterprise dashboard clients
- Implemented role-based permission system with ETS tab gated by VIEW_ETTS role across React frontend and Spring Boot backend
- Built dynamic pricing engine with surge pricing, holiday calendars, and time-based fare rules applied across all trip types — package, outstation, airport, and point-to-point
- Implemented 9 ride-action strategies (assign, reassign, cancel, No Show, OTP, tracking, etc.) via Strategy + Factory design pattern in the enterprise ride management service, enabling zero-change extensibility for new actions
- Developed token-bucket in-memory rate limiter per IP across enterprise backend APIs, preventing abuse and ensuring fair resource allocation under high traffic
- Built cron-driven supply hour validation framework to automatically verify driver shift eligibility before allocation, reducing invalid assignments and improving SLA compliance
- Designed adhoc tripsheet module with garage tracking, split charge fields, client commercials, margin breakdown, and role-gated sections (Ops Reviewer, Client) with full activity log audit trail

Software Development Engineer-II

Oct 2023 – Apr 2026

Oracle, Bangalore, India

Credit Card Approval Platform:

- Scaled RESTful microservices backend automating credit-card approvals using Java, Spring Boot, Hibernate/JPA, SQL, and Kafka, reducing decision latency by **550ms**
- Architected REST API layer aligned to SOLID and OOP principles, integrating **5 microservices** with event-driven architecture
- Integrated a **Gen AI**-assisted risk-flagging layer using LLM-based text classification on applicant profiles to surface high-risk cases, reducing manual review effort by **20%**
- Led 5 interns to deliver Smart Comparator PoC using prompt engineering and OpenAI API integration

JSON Relational Duality Views & Vector AI:

- Reduced CPU consumption by **40%** via active consumer – lazy producer model; developed listener component and Broker APIs

- Managed Oracle Data Guard setup reducing recovery time by **30–50%** via optimized switchover and failover
- Benchmarked JSON workloads up to **5TB/day** to evaluate system limits and influence architecture decisions
- Designed and benchmarked an Embedding-Augmented Generation (EAG) pipeline integrating Oracle 23ai Vector Search with LLM inference APIs, enabling semantic retrieval over large-scale JSON relational data with sub-100ms query latency
- Evaluated Gen AI retrieval strategies (RAG vs. EAG, vector similarity vs. keyword) across **50+ sub-projects**, influencing Oracle's internal architectural guidelines for LLM integration with structured databases
- Automated health checks and replication alerts saving the team **3+ hours** of manual work daily

PROJECTS

SmartLend – Loan Approval Microservices System

2025

- Built event-driven loan approval backend using Java, Spring Boot, Kafka, MySQL, and Docker with 3 independent microservices (Loan, Notification, Audit) communicating via Kafka topics
- Designed REST API with JPA persistence, JUnit tests, Swagger UI, and Dockerized deployment | GitHub

Content Management Web Application

- Automated multi-platform publishing via Instagram and Facebook APIs with secure token-based authentication and Google Drive OAuth2 integration for scheduled media uploads

FidElite – Stock Trading Platform

- Built full-stack trading platform using Angular, Spring Boot, and JPA with real-time portfolio tracking, stock price updates via RESTful APIs, and ACID-compliant Hibernate transaction management

Blockchain-Based Dairy Supply Chain

Apr 2022 – Jun 2022

- Designed Solidity smart contracts to automate validation and payments; conducted 35+ stakeholder interviews translating insights into 12+ MVP features across a decentralized supply chain ledger

LEADERSHIP

Team Lead – Evathon Hackathon | University of Limerick, Ireland

- Ranked **Top 11 out of 350+** global teams; built facial recognition model using ResNet-50 CNN for exam monitoring and attendance during COVID-19

EDUCATION

Sastra Deemed University – B.Tech Computer Science Engineering

Aug 2019 – Aug 2023

GPA: **8.076/10** | Led peer-learning sessions on OS and Data Structures | Coursework: DSA, OS, DBMS